



Fluorescence quantum yield determination of molecules in liquids by thermally driven conical diffraction

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ABSTRACT

We report on a method to determine fluorescence quantum yield (η) of molecules in solution based on the observation of conical diffraction (CD) patterns induced by thermally driven self-phase modulation (SPM). In this approach, the central part of a laser beam is acquired as a function of its power after passing through the sample. If the thermal properties of the solvent are known, one can directly obtain η by measuring the sample optical absorption and the rings produced by CD. Otherwise, if the thermal properties of the sample are unknown, an additional measurement with the same solvent containing nonfluorescent molecules (blue-black ink, for instance) for thermal SPM generation is also needed. In this case, besides determination of η , one can also find the thermal conductivity (K) of the unknown solvent provided that the thermo-optic coefficient (dn/dT) is independently measured. K values obtained with this approach are in good agreement with the literature. CD technique was applied in rhodamine 6 G and a chromophore derivative of vitamin B6 in different solvents. Therefore, the present method may serve as a low-cost optical approach for η determination in liquid samples and for K measurements in unknown solutions.

1. Introduction

Fluorescence quantum yield (η) characterization is essential for the development of novel materials that can be used in light-emitting devices [1,2]. Several techniques based on emission spectra are used for the determination of η relative to a known reference chromophore [3]. Other optical techniques using an integrating sphere, photoacoustic spectroscopy and thermal lens (TL) are currently applied for η determination, but relatively sophisticated alignment or pump-probe configuration are needed [4,5]. Our group introduced the conical diffraction (CD) technique for nonradiative quantum efficiency (φ) determination in liquid samples [6,7]. Now, we present an improved CD procedure using a normalizing reference nonfluorescent sample for η determination. This low-cost technique is simple, and no additional information about thermal conductivity (K) and thermo-optic (dn/dT) coefficients of the samples are required.

In general, the propagation of the intense laser beam through a nonlinear medium leads to several transverse effects [6,8,9]. The spatial profile of the incident beam changes and ring patterns surrounding the

central spot of the ray may be observed at the far-field region. The symmetry of the ring pattern is circular if distortion due to thermal convection is negligible [6,10]. Spatial self-phase modulation (SPM) can be mathematically modeled by using the Fresnel-Kirchhoff diffraction integral [6,11], and different studies of SPM effects are described in various materials [8–12].

The present work applies the normalized CD method for the determination of η in chromophores derivatives of vitamin B6 in different solvents (water, ethanol, chloroform, dimethyl sulfoxide (DMSO) or acetonitrile) and rhodamine 6 G. The vitamin B6 was characterized because their species as pyridoxal, pyridoxine, pyridoxamine and pyridoxal 5'-phosphate have been extensively researched due to their potential benefit to human health, as a key role in inflammatory diseases and antioxidant activity [13,14]. Derivatives of vitamin B6 materials are in continuous development, as they promise for a variety of novel applications as toxic metal ions sensing and pH sensor in aqueous media [15,16].

Also, the present automatized CD setup was used with the non-fluorescent ($\varphi \approx 1$) blue-black ink to determine the thermal

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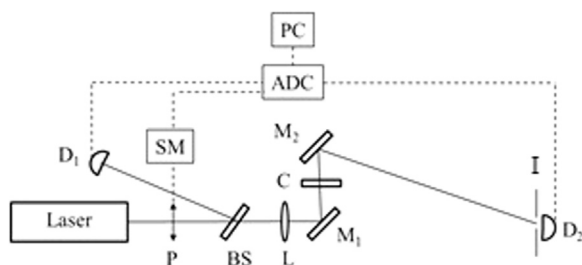


Fig. 1. Experimental setup used to measure CD. Here P is a polarizer, BS a beam splitter, L a lens, M₁ and M₂ are mirrors, C is the cuvette, D₁ and D₂ are detectors, I is an iris, SM is a stepping motor, ADC an analog to digital converter, and PC a personal computer.

conductivities of different solvents. Several techniques are used for thermal conductivity determination in liquids. They include the laser flash method, transient hot-wire method, methods of coaxial cylinders and parallel plates, with typical errors between 1% and 5% [17,18]. For the stationary hot-wire method, 10% errors for K are typical [18]. Since K values of liquid organic and inorganic compounds are sometimes unknown, not available at different temperatures, and the values obtained from the literature by various authors may differ [19,20], we think to be of interest to introduce a new simple technique to measure this parameter.

2. Material and methods

Fig. 1 shows the automatic setup used to measure CD in solutions. The light from a linearly polarized argon ion laser operating at 514.5 nm passes through a rotating linear polarizer that modulates the light intensity according to the Malus law. The polarizer rotation is controlled by a step motor driven by a personal computer interfaced to an analog to digital converter (ADC). In this way, the rotation speed can be chosen to be faster when the intensity varies slowly. A small fraction of the light (~10%) is picked up by a glass plate beam splitter (BS) and directed to a reference detector D_1 connected to the ADC. The incidence of the excitation beam in BS is close to normal (angle less than 10 degrees) and thus the s and p polarizations of reflected light contribute almost equally. This signal provides the input for the x -channel of intensity plots presented later. The voltage measured is converted to incident power by using a power meter just before the sample. After passing through a focusing lens ($f = 20$ cm) the beam is folded vertically by a dielectric mirror (100% reflectivity at the wavelength used, regardless of the polarization) and impinges on a 2-mm thick glass cuvette placed horizontally (to avoid convection effects). CD measurements were performed when the sample is positioned at the focus of the pump beam. CD signal was acquired only in the central part of the ring pattern, and to do so, an iris was placed before detector D_2 that receives light coming from mirror M_2 .

For a given polarizer angle that defines the power incident on the sample, the PC reads and stores the values of the two detectors. Then, the step motor rotates to a new angle, and new values are measured. After acquiring some points enough to cover all power range available, we obtain a curve with several maxima, each one corresponding to a nonlinear phase shift of 2π . Finally, a peak detector routine was used to get the power corresponding to each of the maxima, which are used to yield the relevant thermal parameters.

As we mentioned previously, the thermal conductivity of solvents not yet characterized can be determined if the thermo-optic coefficient is also measured. K can be accomplished if a high accuracy refractometer is employed to measure the refractive index as a function of the temperature, and then taking the derivative. Instead, we chose to use a simpler and accurate technique based on a single arm double interferometer operating at wavelength value (λ) of the 532 nm, with the sample placed in a 2 mm-thick cuvette. This instrument allows the simultaneous measurement of interferences arising from a low-finesse

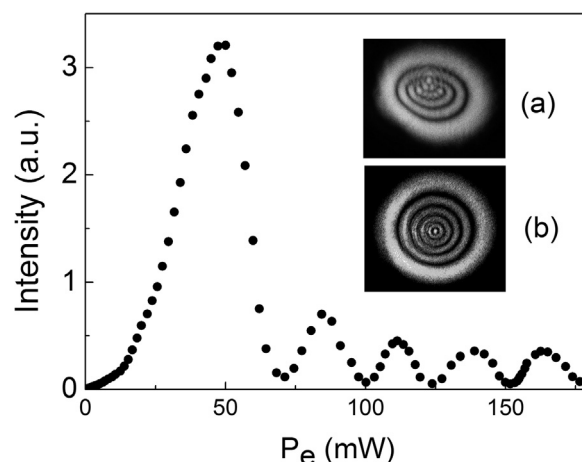


Fig. 2. The intensity of the CD obtained as a function of the incident laser power for blue-black ink diluted in aqueous solution ($\alpha = 4.6 \text{ cm}^{-1}$). The inset shows the ring pattern observed in the far field with sample cuvette at (a) vertical and (b) horizontal positions.

Fabry-Perot etalon and a Mach-Zehnder-type interferometer, in a common-path optical arrangement that makes the device compact and stable [21]. For the measurement of liquid samples, as is the case in the present work, it is necessary to use only the Mach-Zehnder part of the interferometer. In this case, the thermo-optic coefficient can be obtained using $dn/dT = \lambda/(2L \Delta T)$, where ΔT is the temperature spacing between two consecutive fringes and L is the cuvette thickness. Fluorescence spectra were obtained with a portable spectrometer (Ocean Optics USB2000+) by pumping the sample at 532 nm. The absorbance spectra were acquired by using a halogen lamp and the portable spectrometer.

Rhodamine 6 G and blue-black ink were purchased from Acros Organics and Sheaffer, respectively. Pyridoxal 5'-phosphate hydrate (vitamin B6) was obtained from Sigma Aldrich. The derivative of vitamin B6 was synthesized as described in reference [15].

3. Results and discussion

The intensity of the central part of the laser beam obtained with the CD setup as a function of the laser power (P_e) is shown in Fig. 2 for blue-black ink in aqueous solution. The insets in this figure show typical ring patterns obtained when the cuvette was placed horizontally and vertically. For a cuvette positioned vertically, the far field pattern observed consists of a set of distorted rings that are not concentric (Fig. 2a). This effect can be ascribed to the convection flow of the solution inside the cuvette, which may lead to distortions of the transmitted wavefront [10]. Otherwise, for a horizontal cuvette with a short path length, any convection can be neglected and the rings observed at the far field position are well concentric (Fig. 2b). Therefore, for the sake of automating the measurement, the CD setup used the horizontal configuration of the cuvette.

According to the general CD theory, the number of rings, N , is proportional to the nonlinear phase change, $\Delta\phi_{NL}$ [22,23], which in the case of thermal effects [24] is proportional to the power of the excitation laser beam, P_e . Therefore, $N = \beta P_e$ where [6,7,25]:

$$\beta = \frac{\alpha \varphi L_{\text{eff}}}{2\pi K \lambda} \left(\frac{dn}{dT} \right) \quad (1)$$

and $\alpha \text{ (cm}^{-1}\text{)}$ is the optical absorption coefficient at the excitation wavelength (λ), $L_{\text{eff}} = (1 - \exp(-\alpha L))/\alpha$ is the effective length, $L \text{ (cm)}$ is the sample thickness. φ called the absolute nonradiative quantum efficiency represents the fraction of the absorbed energy converted into heat and K is the thermal conductivity ($\text{W cm}^{-1} \text{K}^{-1}$). In this case, η can be determined according to the expression:

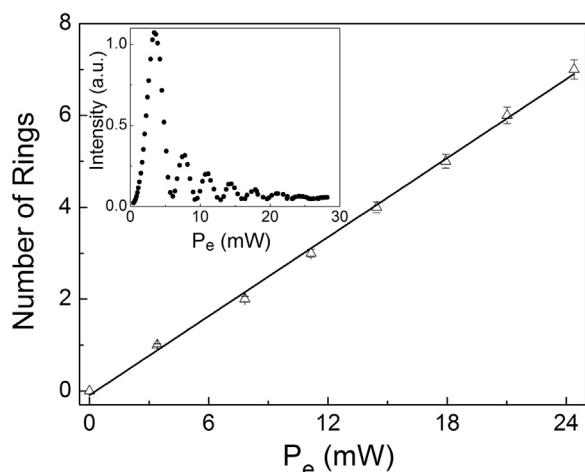


Fig. 3. The number of rings N as a function of the laser power for R6G diluted in chloroform open triangles and linear fit with a slope $\beta = (287 \pm 5) \text{ W}^{-1}$ for $L = 2 \text{ mm}$ and $\lambda = 514 \text{ nm}$.

$$\eta = (1 - \varphi) < \lambda_{em} > / \lambda \quad (2)$$

where $< \lambda_{em} >$ is the average emission wavelength.

Fig. 3 shows the linear behavior of the number of rings (N) as a function of the P_e for rhodamine 6 G (R6G) in chloroform. R6G in ethanol solution was also measured, but the experimental data are not shown here. The result for chloroform was adjusted with a linear function and β was determined from the slope of the fit. Averaged β values were obtained considering three independent fresh samples for each solution investigated. From β using Eq. (1) and K given in Table 1 [26–28], the parameter $|\varphi dn/dT| = (1.80 \pm 0.06)$ and $(0.50 \pm 0.01) \times 10^{-4} \text{ K}^{-1}$ were determined for R6G in chloroform and ethanol solutions, respectively. φ was determined using the value of dn/dT obtained from the interferometric measurements (Table 1). Finally, η was evaluated from Eq. (2). It is important to emphasize that dn/dT results are in good agreement with those reported elsewhere [29–31]. Besides, $< \lambda_{em} > = 591 \text{ nm}$ (572 nm) was evaluated from the fluorescence spectra for R6G in chloroform (ethanol solution). The quantum yield η average results were $\eta = (0.83 \pm 0.03)$ and (0.98 ± 0.03) for R6G in chloroform and ethanol solution, respectively. The results are in good agreement with those of the literature [32,33], showing that our setup is reliable for quantum yield determination.

Since we have a straightforward and accurate interferometric setup [21] to measure dn/dT , we can also use the CD technique to determine the thermal conductivity of liquid samples. For this purpose, CD measurements were carried out in solutions of blue-black ink diluted in different solvents. For ink in aqueous solution with $\alpha = (4.6 \pm 0.1) \text{ cm}^{-1}$, using dn/dT from Table 1 and considering $\varphi \approx 1$ (i.e. $\eta \approx 0$), the average value of the β and K obtained were $(29 \pm 1) \text{ W}^{-1}$ and $(0.59 \pm 0.03) \text{ W/mK}$, respectively. The K value found is in good agreement with that from the literature (Table 1, error $\sim 2\%$).

The above result indicates that the CD technique applied to

Table 1
The average values for thermo-optic coefficients in different solvents.

Solvent	K (W/mK)	$-dn/dT$ (10^{-4} K^{-1})		
		532 nm (22 °C)	Literature	Err. (%)
acetone	0.1605 [26]	5.1 ± 0.2	4.9–5.4 [29]	1
acetonitrile	0.2034 [27]	4.7 ± 0.2	4.6 [29]	2
chloroform	0.1175 [26]	6.4 ± 0.3	6.545 [30]	2
DMSO	0.223 [28]	4.8 ± 0.2	4.9 [31]	2
ethanol	0.1635 [27]	4.2 ± 0.1	3.6–4.0 [29]	10
water	0.5981 [27]	0.92 ± 0.03	0.8–1.0 [29]	2

Table 2

Results obtained with automated CD for ink nonfluorescent sample.

Sample	$\alpha \text{ cm}^{-1}$ (514.5 nm)	αL_{eff}	$\beta \text{ (W}^{-1}\text{)}$	$K \text{ (W/mK)}$	K Err. (%)
acetone	4.41 ± 0.05	0.59 ± 0.01	527 ± 8	0.177	9
acetonitrile	2.18 ± 0.04	0.35 ± 0.01	239 ± 5	0.213	5
DMSO	3.40 ± 0.05	0.49 ± 0.01	305 ± 7	0.239	7
ethanol	4.0 ± 0.1	0.55 ± 0.01	406 ± 20	0.176	7
water	4.6 ± 0.1	0.60 ± 0.01	29 ± 1	0.589	2

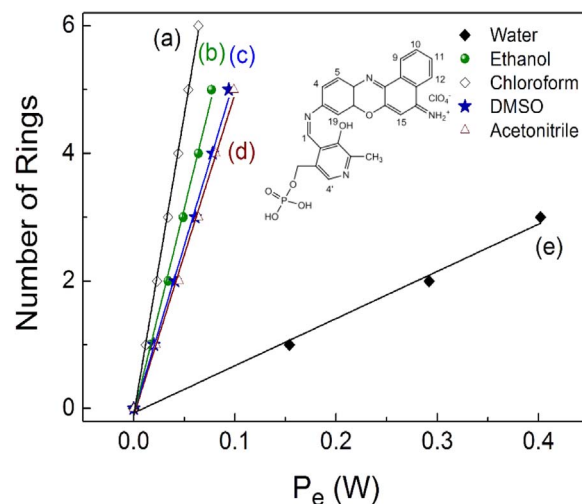


Fig. 4. CD results for vitamin B6 diluted in different solvents (a) chloroform, (b) ethanol, (c) DMSO, (d) acetonitrile and (e) aqueous solutions. The solid lines were obtained using a linear fitting, and β parameters are presented in Table 3. The inset shows the chemical structure of the B6 derivative used.

nonfluorescent samples is advantageous for the determination of K parameter. Thermal conductivity was achieved without a complicated experimental setup and points out future applications of this technique. The CD results for blue-black ink diluted in others solvents are presented in Table 2, and K values are also in good agreement with those found in the literature [26–28]. However, we must point out that the accuracy of the results depends on the cuvette position, meaning that it must be located as close to the focus as possible. For a displacement of 1 mm in the cuvette position concerning the laser waist, the φ error is $\sim 7\%$.

Fig. 4 presents the CD results for a derivative chromophore of vitamin B6 in different solvents (acetonitrile, chloroform, DMSO, ethanol, and water). The slopes β obtained with linear fits are presented in Table 3. Using the values of $< \lambda_{em} >$ that were determined from the fluorescence spectra, the thermo-optical and thermal conductivity parameters (Table 1), η values were calculated for each liquid sample. Average η values are also presented in Table 3. Relative high quantum efficiencies were determined, in agreement with another B6 derivative [15]. η of B6 derivative composite in miliQ water is reported as being 0.51, using cresyl violet in ethanol as reference [15]. This result is in agreement with that presented in Table 3 for B6 in aqueous solution. The number of rings (N) as a function of laser beam power is shown in Fig. 5 for B6 diluted in chloroform in different concentrations ($\alpha = 0.2\text{--}1.2 \text{ cm}^{-1}$), and η values change 3% in this concentration range.

The above results were obtained with known thermo-optic and thermal conductivities values given in Table 1. In many cases, however, the sample is suspended in a pure solvent or mixture of solvents whose thermal properties are unknown. To circumvent this problem we propose an implementation of the CD technique using a nonfluorescent molecule ($\varphi \approx 1$) to remove the dependence of the parameters dn/dT and K of the unknown solvent. This nonfluorescent normalization sample method will be referred to as normalized CD technique (NCD),

Table 3
Results obtained with automated CD for vitamin B6 derivative.

Sample	$\alpha \text{ cm}^{-1}$ (514.5 nm)	$\alpha_{\text{L-eff}}$	$\langle \lambda_{\text{em}} \rangle$ (nm)	β (W^{-1})	η	η^a
acetonitrile	1.22 ± 0.04	0.22 ± 0.01	634	51 ± 1	0.86 ± 0.01	0.81
chloroform	1.2 ± 0.1	0.21 ± 0.01	636	94 ± 5	0.91 ± 0.03	–
DMSO	1.20 ± 0.04	0.21 ± 0.01	643	56 ± 1	0.75 ± 0.02	0.71
ethanol	1.3 ± 0.1	0.23 ± 0.01	644	64 ± 4	0.81 ± 0.03	0.78
water	1.8 ± 0.1	0.30 ± 0.01	642	7.4 ± 0.4	0.59 ± 0.04	0.61

^a Using NCD.

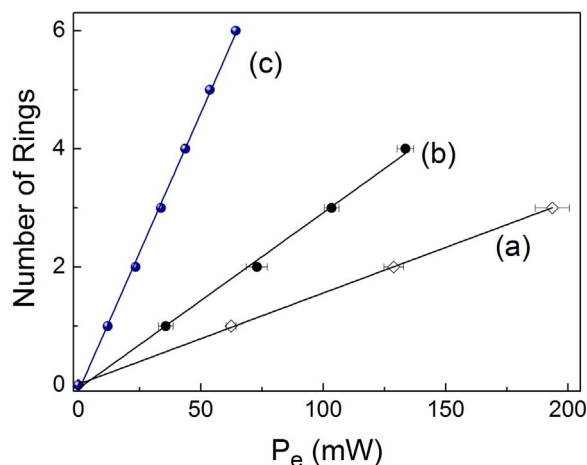


Fig. 5. CD results for vitamin B6 diluted in chloroform at different concentrations: (a) 0.2, (b) 0.41 and (c) 1.21 cm^{-1} at 514.5 nm. The solid lines were obtained using a linear fitting, and β parameters obtained are (a) (15.5 ± 0.1) , (b) (29.7 ± 0.8) and (c) (94 ± 5) W^{-1} , respectively.

for short. In this approach, using the number of rings obtained for the two samples (fluorescent and nonfluorescent) and Eq. (1), one obtains:

$$\varphi = \frac{(\alpha L_{\text{eff}})_{\text{ref}} \beta_{\text{fluor}}}{(\alpha L_{\text{eff}})_{\text{fluor}} \beta_{\text{ref}}} \quad (3)$$

where β_{ref} is the slope of N versus P_e curve for ink diluted in the same solvent as used for the fluorescent molecule, whose slope is β_{fluor} . Eqs. (2) and (3) allow us to obtain η results that are presented in Table 3 for the derivative chromophore of vitamin B in different solvents (acetonitrile, DMSO, ethanol and water). The NCD results are in good agreement with that previously found with the conventional CD technique. These results points out that the CD technique applying the nonfluorescent normalization sample is a useful and fast method for η determination, where no information about thermo-optic and thermal conductivity parameters from solvent are required.

4. Conclusions

The present work introduces an automated setup to measure conical diffraction (CD) in solutions for fluorescence quantum yield (η) determination. An important feature is a cuvette placed in a horizontal position that avoids convection flows and provides good quality far-field patterns. The CD technique was applied to blue-black ink, as a nonfluorescent molecule for thermal conductivity (K) determination in liquids. It was also applied to a solution of the rhodamine 6 G (R6G), and η results obtained are in good agreement with the literature. A method using the normalization to a reference nonfluorescent sample was proposed (NCD) to avoid the knowledge of the thermo-optic (dn/dT), and K . Relative high values of η determined for B6 derivatives in different solvents is in agreement with values obtained for other B6 derivatives. The NCD technique stands out as the simplest method for quantum yield determination.

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Contributors

SR De L was responsible for CD experimental measurements in derivative of vitamin B6 and ink in different solvents, spectroscopic of the solutions, the single arm double interferometer measurements and average results, and analysis. GJP was responsible for the test of the automatized CD, and measurements in R6G in different solvents. DNM and AAA were responsible for the interpretation of the results and discussions. EO and CL were responsible for the synthesis of B6 derivative, and both have collaborated with paper writing and reading. SCZ performed CD automation and contributed to the paper writing. VP proposed CD and NCD methods, paper writing, analysis, and discussions.

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